

ECN 712 Fall 2025
EE Schlee Problem Set V
Due 25 September

1. Let $f : C \times T \rightarrow \mathbb{R}$, where T is a convex subset of $\mathbb{R}^m$ for some positive integer m and C is an arbitrary nonempty set. Suppose that there is a maximizer of $f(\cdot, t)$ on C for every $t \in T$ and define $V(t) = \max_{x \in C} f(x, t)$ for every $t \in T$. Prove that if $f(x, \cdot)$ is convex on T for every $x \in C$, then V is convex: in words, *the maximum of a family of convex functions is convex*. (Corollaries: on any convex set P of prices for which it is defined, the profit function $\pi(p) = \max_{y \in Y} p \cdot y$ is convex; and the expenditure and cost functions are concave in prices.)
2. Prove this *generalized law of supply and demand* (Debreu, 1959, p. 47),

$$\Delta p \cdot \Delta y \geq 0,$$

where $\Delta p = (p^1 - p^0)$, $\Delta y = (y^1 - y^0)$, and, for $t = 0, 1$, the plan y^t maximizes a firm's profit on a production set Y at $p^t \in \mathbb{R}_{++}^L$. Interpret the inequality for the general case and for the case in which only one price changes.

3. A firm uses labor and other inputs $x \in \mathbb{R}_+^m$ to produce a single output, q , according to a production function $q = f(n, h, x)$, where n is the number of employees and h the number of hours each employee works (i.e. it may make a difference whether one worker works 80 hours or two work 40 hours each). If either $n = 0$ or $h = 0$, then $f(x, n, h) = 0$ (labor is an essential input). It sells its output in a competitive market at price $p \in \mathbb{R}_{++}$. Prices for the m non-labor inputs are $r \in \mathbb{R}_{++}^m$. The firm pays each worker a positive hourly wage rate w , and also provides a health care benefit costing it $\$b > 0$ per employee. (a) What is the effect of an increase in the health-care benefit on the number of employees? (b) What is the effect of an increase in the wage rate on the total number of hours worked? [Note: there are at least *four* comparative statics arguments that give the conclusion, but some require additional assumptions. If the argument you chose requires additional assumptions, be explicit about them.]
4. A firm produces an output q using $n > 1$ inputs according to a continuous, strictly increasing, strictly supermodular, production function f on $\mathbb{R}_+^n$. (Equivalently, it satisfies pairwise strictly increasing differences.) The firm sells its output at a price of $p > 0$ per unit and buys input i at a price of $w_i > 0$ per unit. Suppose that the price of input 1 *decreases* from w_1^0 to $w_1^1 > 0$. Let the input vector z^t maximize the firm's profit when $w_1 = w_1^t$ for $t = 0, 1$.
 - (a) Explain why you cannot simply invoke Topkis's theorem (in the Monotone Comparative Statics handout) to assert that $z^1 \geq z^0$.
 - (b) Suppose that, for $t = 0, 1$, the point $\underline{z}^t$ is the *smallest* solution to the firm's problem at w_1^t in the sense that if z' is a solution at w_1^t , then $\underline{z}^t \leq z'$. Prove that $\underline{z}^1 \geq \underline{z}^0$. [Suggestion: there are two cases, $\underline{z}_1^0 = \underline{z}_1^1$ and $\underline{z}_1^0 \neq \underline{z}_1^1$. You might start with the $n = 2$ case.]¹ I do not ask you to do it here, but an analogous assertion holds for any *largest* input demand, if they exist.

¹Part (b) relies on the *existence* of smallest solutions. I'll sketch an argument for their existence here. Fix $(p, w) \gg 0$ and let $Z^* \neq \emptyset$ be the set of solutions. Use SSPM of f to show that any two solutions are comparable under the " $\geq$ " relation on $\mathbb{R}_+^n$; and then use that the feasible choices for z are bounded below (by 0) and that the firm's profit, $pf(z) - w \cdot z$, is continuous on $\mathbb{R}_+^n$. A smallest choice will be the coordinatewise infimum of the set of solutions Z^* . (It does not follow that there exist largest solutions, since the set of inputs is not bounded above.)